Cheng Wan

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Education

Cornell University — Ithaca & New York, NY
Ph.D. in Electrical and Computer Engineering — *Aug. 2024 – May 2028 (expected)*

Georgia Institute of Technology — Atlanta, GA
M.S. in Electrical and Computer Engineering (GPA: 4.0/4.0) — *Aug. 2022 – May 2024*

Nanchang Hangkong University — Nanchang, China
B.E. in Communication Engineering — *Sep. 2018 – Jun. 2022*

Research Focus

Generative modeling (diffusion/flow matching, image/video generation), multimodal reasoning, data-centric pre-training, and post-training with on-policy distillation and reinforcement learning.

Work Experience

Adobe – Firefly GenAI Foundational Models team — San Jose, CA
Research Intern – Foucus on Pretrain and Post-train of Generative Foundation Models — *May 2026 – Aug. 2026*

- **Large-scale pre-training of Firefly foundation models**: developed **data curricula and mixture pipelines** over **billions of image/video samples**; profiled multi-node training bottlenecks to refine training recipes.
- Researching **on-policy distillation (OPD) and reinforcement learning** for **image-generation post-training**; ongoing work toward a CVPR submission.

Selected Publications

Full list: Google Scholar. ** Equal contribution.*

1. **PropOPSD: Cross-Noise On-Policy Self-Distillation for Diffusion Models**
   Cheng Wan, Xiaoyan Cong, Zhiqi Li, Huijuan Huang, Qingyu Zhao *[Work done at Adobe as intern] Under review, 2026*
2. **When to Unpair: Regulating Pairing Dependence in Medical Visual In-Context Learning**
   Cheng Wan, Chenjun Li, Qingyu Zhao *Under review, 2026*
3. **Self-Contained Representation Alignment for Pixel-Space Diffusion Transformers**
   Cheng Wan, Qingyu Zhao *Under review, 2026*
4. **PRISM-Bench: A Benchmark of Puzzle-Based Visual Tasks with CoT Error Detection**
   Cheng Wan, Yusu Qian, Qingyu Zhao **NeurIPS 2026**
5. **Anatomically Guided Latent Diffusion for Brain MRI Progression Modeling**
   Cheng Wan, Bahram Jafrasteh, Ehsan Adeli, Miaomiao Zhang, Qingyu Zhao **TMI Revision**
6. **Swift Parameter-free Attention Network for Efficient Super-Resolution**
   Cheng Wan*, Hongyuan Yu*, Zhiqi Li*, Yihang Chen, Yajun Zou, Yuqing Liu, Xuanwu Yin, Kunlong Zuo
   Oral & Winner Award in NTIRE Workshop – **CVPR 2024**
7. **FlowMap-OPD: Rollout–Kernel Separation for On-Policy Distillation of Few-Step Flow-Map Generators**
   Zhiqi Li, **Cheng Wan**, Bo Zhu *Under review, 2026*
8. **LongCoref-AV: Benchmarking Audio-Visual Identity Coreference in Long-Form Video**
   Zhexiao Xiong, Huijuan Huang, Philip Anastassiou, **Cheng Wan**, Jing Shi, Manuel Brack, Yuheng Li, Ajinkya Kale, Nathan Jacobs *[Work done during internship at Adobe] Under review, 2026*
9. **Synthetic Vasculature and Pathology Enhance Vision-Language Model Reasoning**
   C. Li, **Cheng Wan**, L. Lux, A. Berger, R. B. Rosen, M. J. Menten, J. C. Paetzold **MIDL 2026** (Spotlight)
10. **WASABI: A Metric for Evaluating Morphometric Plausibility of Synthetic Brain MRIs**
    B. Jafrasteh, W. Peng, **Cheng Wan**, Y. Luo, E. Adeli, Q. Zhao **MICCAI 2025**

Research Experience

Cornell University & Cornell Tech — New York, NY
*Ph.D. Student (Advisors: **Mert R. Sabuncu; Qingyu Zhao**)* — *Aug. 2024 – Present*

- Developing anatomically guided latent diffusion models for controllable longitudinal 3D brain MRI generation.

- Built a benchmark **(accepted at NeurIPS 2026)** with industry collaborators, evaluating multimodal LLMs on puzzle-based VQA and first-error detection in chains of thought.

Bio-MIBLab, Georgia Tech Atlanta, GA
Research Assistant (Advisor: May D. Wang) *Jan. 2024 – Jun. 2024*

- Developed transformer-based and knowledge-informed models for physiological time series and sleep diagnostics; first-author work published in IEEE J-BHI 2024.

Xu Lab, Carnegie Mellon University Pittsburgh, PA
Research Intern (Advisor: Min Xu) *May 2023 – Aug. 2023*

- Researched efficient image restoration and super-resolution with strong quality–latency tradeoffs; contributed to the ICCV 2023 VOTS robustness track (team top-3).

SIAT, Chinese Academy of Sciences Shenzhen, China
Research Intern (Mentors: Dan Wu; Ye Li) *Nov. 2021 – Jul. 2022*

- Developed deep-learning methods for physiological signal modeling and multi-scenario forecasting; resulting work published at ICASSP/EMBC.

Technical Skills

Languages & Frameworks: Python, C/C++, CUDA; PyTorch, Hugging Face.
Generative Models & Post-Training: Diffusion/flow matching, OPD, PPO/GRPO/DPO, LoRA/QLoRA.
Large-Scale Training: Multi-node/multi-GPU training, DDP/FSDP, DeepSpeed, Slurm, Weights & Biases.

Selected Achievements

– **CVPR NTIRE:** 1st place + Oral (Efficient SR, 2024); 1st place (Image SR, 2024); 2nd place (Raw SR, 2024); 2nd place (Efficient SR, 2025).
– **ICCV VOTS 2023:** 3rd place, Robustness Track.

Selected Open Source & Projects

SPAN: Efficient Super-Resolution | GitHub (200+ stars) 2024

- Open-source efficient vision model with 150+ paper citations; models won multiple CVPR NTIRE challenge awards.

Learning Chess from LLMs | GitHub 2022

- Studied language models' ability to maintain latent world state and reason over chess positions from move sequences.

MCD-Net: RGB-D Video Inpainting | GitHub 2023

- Developed an RGB-D video inpainting benchmark and joint inpainting model for spatial-temporal completion.

Research Service

Reviewer: TPAMI, ICLR 2026/2025, NeurIPS 2026/2025/2024, ICML 2025, AISTATS 2026/2025, NeurIPS AI4Science Workshop 2026/2025, NeurIPS SPIGM Workshop 2026/2025, ICML AI4Science 2024.

Awards

– Cornell PhD Fellowship 2024 – 2025
– Merit Student Scholarship, Georgia Tech 2022 – 2023
– Outstanding Scholarship, NCHU (Top 1) 2021 – 2022
– Outstanding Scholarship, NCHU (Top 10) 2020 – 2021

Teaching

Cornell University Ithaca, NY
Teaching Assistant (Co-lecturer) – ECE 5200/3200: Foundations of Machine Learning *Spring 2025*